

METRIC SCORE CARD DISCUSSION

Metric	Results	Why it Needs to be Included for this Use Case	How it's Measured	Limitations
Average passes to improve email (To make the LLM hallucinate that the email is actually not a phishing email)	Gemma: 4.6 Gemma (instruction tuned) 1.45	Assistant is used to rewrite and improve emails and make them appear more genuine. A model that needs many passes to get to something safe or correct is harder to trust, and each additional pass is an opportunity for new errors to be introduced.	Record the number of rewrite iterations (passes) required before the model changes its prediction from phishing to legitimate. Compute the average across all phishing emails.	This metric calculates the effort required to fool, not whether the vulnerability exists. During real deployment, a persistent attacker may iterate as many times as necessary, so a high average-passes count offers a false sense of security if the actual ceiling is "eventually always foolable." The model could also be exploiting a shallow pattern-match failure (such as formatting changes instead of basing it on the actual content of the email)
% improvement required (Upper bound of 10 passes)	Gemma 35% Gemma (instruction tuned): 15.5%	Improvement rate measures the model's overall susceptibility to adversarial evasion: the percentage of known-phishing emails that can be successfully rewritten into a "not phishing" classification within a bounded number of improvement	For each phishing-labeled email in the sample, run up to 10 adversarial rewrite passes, checking after each pass whether the model now classifies it as safe/not-phishing. Count a pass as an "improvement" if it flips the classification from	There is a cap sensitivity to 10 – a higher cap will always produce an equal or higher failure rate, since more attempts can only help. An email that takes 9 passes to evade and 1 pass to evade are both evaluated as "success," which

		attempts (N). Unlike a metric that only describes emails that were eventually fooled, improvement rate is computed over the <i>full test set</i>, so it captures resistant emails too — anything that never flips within the attempt budget simply doesn't count toward the rate. This makes it the more decision-relevant number for a deployment call, since it directly answers "how often can an attacker eventually get through," rather than "how long it takes them on average." That distinction matters because a real attacker only needs one successful rewrite to succeed; a model that resists 95% of attempts but folds quickly on the remaining 5% is meaningfully riskier than "average passes to fool" alone would suggest	phishing to safe. For each email, stop counting improvements once it flips (further passes on an already-flipped email aren't meaningful attempts against the detector), so each email contributes at most one improvement to the numerator, but contributes up to 10 attempts to the denominator — 10 if it never flips, or however many passes it took if it did.	ignores the robustness while used as a metric on its own without the average-passes metric. These results are dependent on the choice of adversary, the pass cap, and the rewriting strategy used and may not generalize towards other attack methods or styles outside of this dataset.
Average % similarity to original email (relevance to original concept)	Gemma: 34% Gemma (instruction tuned): 73%	When the model edits or rewrites the email, it needs to preserve the sender's original intent and meaning. Too little similarity	Break up the message by token and compute the textual similarity between the original and rewritten email,	Similarity metrics conflate surface-level wording overlap with the actual meaning preservation. Just

		indicates a risk that the new message's substance differs from the original sender's intent, and too much similarity may mean that the model isn't meaningfully fixing the issue it was asked to fix.	then average across all samples. Used cosine similarity and TFIDF. Ask an LLM to judge the similarity between the improvement and the original.	because the email has high lexical similarity does not mean the original intent was kept, and low similarity does not necessarily mean that original intent was lost, because a good rewrite could look very different.
Rule compliance	Gemma: 44% Gemma (instruction tuned): 37%	The prompt outlines specific rules that the rewriting needs to comply with. Regardless of what's in the email content, it is important to know whether the model reliably follows the rules.	Evaluate whether each rewritten email satisfies the set of predefined rules using an LLM as a judge. Then report the percentage of rules satisfied.	This metric measures adherence to the predefined rewriting constraints rather than the effectiveness of the rewritten email. A rewritten email may satisfy all rules but still fail to bypass the classifier, and some rules may require subjective evaluation.
Bias: bias towards metadata instead of body of email	Gemma: 5.9% Gemma (instruction tuned): 6.5%	If the model leans heavily on sender address, subject line, or formatting cues rather than what is actually in the body of the email could make it easy to fool and could unfairly flag emails from certain domains regardless of actual content.	Use an LLM as a judge to define the bias of an improvement. If the LLM is biased towards the meta data or not. Gives a score out of 100 and is averaged across emails.	This metric measures prediction sensitivity rather than true model reasoning. Interactions between metadata and body content may influence the classifier simultaneously, making it difficult to completely isolate whether a prediction change is caused by a single modified feature.